Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Jordao Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
6025 S Central AVE	Turlock	Stanislaus	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 04/01/2007

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X030-X013-XXXX X045-X030-X022-XXXX X057-X004-X013-XXXX X057-X004-X014-XXXX

B. OPERATORS

Jordao, Joe

Operator name: Jordao, Joe

Telephone no.: (209) 656-8438 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
6025 S Central AVE	Turlock	CA	95380

This operator is responsible for paying permit fees.

C. OWNERS

Jordao, Joe

Legal owner name: Jordao, Joe

Telephone no.: (209) 656-8438 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
6025 S Central AVE	Turlock	CA	95380

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	252	398	173	152
Number under roof	873	88	0	0	0	0
Maximum number	873	88	252	398	173	152
Average number	873	88	252	398	173	152
Avg live weight (lbs)	1,400	1,450	900	600		

Predominant milk cow breed: Holstein

Average milk production: 73 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 31,304.73 *tons per reporting period*

Total nitrogen from manure: 391,158.08 *lbs per reporting period*

After ammonia losses (30% loss applied): 273,810.66 *lbs per reporting period*

Total phosphorus from manure: 64,039.29 *lbs per reporting period*

Total potassium from manure: 168,908.24 *lbs per reporting period*

Total salt from manure: 431,287.65 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 11,727,102 *gallons*

Total nitrogen generated: 56,722.41 *lbs*

Total phosphorus generated: 9,459.74 *lbs*

Total potassium generated: 89,864.38 *lbs*

Total salt generated: 569,163.56 *lbs*

	11,727,102	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	11,727,102	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - DW-5627 Central	Ground water
Domestic Well - DW-6025 Central	Ground water
Irrigation Water - Canal	Surface water
Irrigation water - River Water	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	37	37	2	process wastewater	X057-X004-X014-XXXX
2	37	37	2	process wastewater	X057-X004-X014-XXXX
3	39	39	2	process wastewater	X057-X004-X013-XXXX
4	98	98	2	manure	X045-X030-X013-XXXX
5	75	75	2	manure	X045-X030-X022-XXXX
Carpenter	260	260	2	manure	X058-X023-X002-XXXX
Crowslanding	71	71	6	manure	X057-X001-X010-XXXX
Hogan South	150	150	3	manure	X057-X025-X010-XXXX
Hogan West	100	100	0	none	X057-X026-X009-XXXX
The 65	65	65	2	manure	X045-X002-X037-XXXX X045-X004-X047-XXXX
Totals for areas that were used for application	832	832	23		
Totals for areas that were not used for application	100	100	0		
Land application area totals	932	932	23		

B. CROPS AND HARVESTS

1

Field name: 1

11/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 11/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/19/2023	572.00 *ton*	Dry-weight		64.2	11,800.00	2,700.00	19,500.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.46	130.61	29.89	215.84	1,117.97

1

05/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 05/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/29/2023	1,075.66 *ton*	Dry-weight		64.6	13,200.00	2,700.00	10,500.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.07	271.69	55.57	216.12	1,152.64

2

Field name: 2

11/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 11/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/19/2023	596.00 *ton*	Dry-weight		64.5	12,100.00	2,600.00	19,400.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.11	138.38	29.74	221.87	1,166.55

05/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 05/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/29/2023	1,021.00 *ton*	Dry-weight		64.4	13,900.00	2,700.00	10,900.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.59	273.10	53.05	214.16	1,100.25

3

Field name: 3

11/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 39 Plant date: 11/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/19/2023	591.00 *ton*	Dry-weight		65.1	12,500.00	2,700.00	20,200.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.15	132.22	28.56	213.66	1,068.32

05/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 05/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/29/2023	1,023.00 *ton*	Dry-weight		64.9	13,000.00	2,600.00	10,900.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.23	239.38	47.88	200.71	994.36

4

Field name: 4

11/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 98 Plant date: 11/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/19/2023	1,251.93 *ton*	Dry-weight		64.3	12,100.00	2,500.00	18,900.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.77	110.37	22.80	172.39	893.88

4

05/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 98 Plant date: 05/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/29/2023	2,501.00 *ton*	Dry-weight		64.3	13,800.00	2,800.00	11,000.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	48.00	165.00	0.00
Total actual harvest content	25.52	251.46	51.02	200.44	983.96

5

Field name: 5

11/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 75 Plant date: 11/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/19/2023	1,299.00 *ton*	Dry-weight		66.0	11,000.00	2,600.00	19,800.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.32	129.55	30.62	233.20	1,177.76

05/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 05/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/29/2023	1,901.00 *ton*	Dry-weight		62.0	14,200.00	2,700.00	9,800.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.35	273.54	52.01	188.78	1,098.02

Carpenter

Field name: Carpenter

11/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 260 Plant date: 11/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/19/2023	3,401.00 *ton*	Dry-weight		63.8	12,300.00	2,700.00	20,100.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.08	116.49	25.57	190.36	975.46

05/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 260 Plant date: 05/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/29/2023	7,002.00 *ton*	Dry-weight		65.7	13,700.00	2,800.00	9,900.00		3.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.93	253.10	51.73	182.90	554.24

Crowslanding

Field name: Crowslanding

Crowslanding

11/10/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 71 Plant date: 11/10/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/05/2023	83.00 *ton*	Dry-weight		9.2	31,600.00	4,000.00	25,800.00		9.80
05/10/2023	81.00 *ton*	Dry-weight		9.2	31,600.00	4,000.00	25,800.00		9.80
06/12/2023	77.00 *ton*	Dry-weight		9.2	31,600.00	4,000.00	25,800.00		9.80
07/18/2023	81.00 *ton*	Dry-weight		9.2	31,600.00	4,000.00	25,800.00		9.80
08/19/2023	81.00 *ton*	Dry-weight		9.2	31,600.00	4,000.00	25,800.00		9.80
09/20/2023	82.00 *ton*	Dry-weight		9.2	31,600.00	4,000.00	25,800.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	6.83	392.00	49.62	320.05	1,215.70

Hogan South

Field name: Hogan South

11/05/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 150 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/05/2023	366.14 *ton*	Dry-weight		5.3	24,300.00	3,500.00	16,600.00		8.40
05/10/2023	197.19 *ton*	Dry-weight		5.3	24,300.00	3,500.00	16,600.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	3.76	172.85	24.90	118.08	597.49

Hogan South

06/20/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 150 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	1,510.00 *ton*	Dry-weight		83.2	19,500.00	4,200.00	32,900.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	10.07	65.96	14.21	111.28	368.68

The 65

Field name: The 65

11/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 65 Plant date: 11/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/19/2023	912.00 *ton*	Dry-weight		65.6	11,700.00	2,600.00	19,700.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.03	112.94	25.10	190.17	984.62

05/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 65 Plant date: 05/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/29/2023	1,723.00 *ton*	Dry-weight		66.0	13,800.00	2,700.00	11,000.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.51	248.75	48.67	198.28	1,045.46

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/28/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 11/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.99	14.60	135.21	700.39	599,492.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.45	5,031,409.00 *gal*
Application event totals		85.99	14.60	135.21	737.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	79.93	14.72	138.94	935.45	609,939.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.51	5,174,635.00 *gal*
Application event totals		79.93	14.72	138.94	973.96	

1 - 05/04/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 05/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.27	5,544,818.00 *gal*
Application event totals		0.00	0.00	0.00	41.27	

1 - 05/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.78	5,478,808.00 *gal*
Application event totals		0.00	0.00	0.00	40.78	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.56	17.16	162.29	1,138.86	901,692.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.53	5,445,803.00 *gal*
Application event totals		90.56	17.16	162.29	1,179.39	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	120.31	22.16	209.14	1,408.05	918,090.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.70	5,736,762.00 *gal*
Application event totals		120.31	22.16	209.14	1,450.75	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	136.35	25.11	237.02	1,595.79	1,040,502.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.19	5,668,058.00 *gal*
Application event totals		136.35	25.11	237.02	1,637.98	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.46	5,839,817.00 *gal*
Application event totals		0.00	0.00	0.00	43.46	

1 - 05/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.23	5,942,873.00 *gal*
Application event totals		0.00	0.00	0.00	44.23	

2 - 11/28/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough Plant date: 11/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	108.34	10.85	111.80	623.34	588,035.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	27.25	3,661,550.00 *gal*
Application event totals		108.34	10.85	111.80	650.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	65.44	12.05	113.75	765.86	499,361.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	27.60	3,708,493.00 *gal*
Application event totals		65.44	12.05	113.75	793.46	

2 - 05/04/2023: Corn, silage

Field name: 2

Crop: Corn, silage Plant date: 05/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

2 - 05/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.85	5,488,809.00 *gal*
Application event totals		0.00	0.00	0.00	40.85	

06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.34	5,554,152.00 *gal*
Application event totals		0.00	0.00	0.00	41.34	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	128.04	21.74	201.31	1,042.81	892,584.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.37	5,423,467.00 *gal*
Application event totals		128.04	21.74	201.31	1,083.17	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	122.91	20.87	193.26	1,001.10	856,884.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.19	5,668,058.00 *gal*
Application event totals		122.91	20.87	193.26	1,043.28	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.21	17.48	165.24	1,159.57	918,090.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.93	5,633,706.00 *gal*
Application event totals		92.21	17.48	165.24	1,201.50	

2 - 05/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.70	5,736,762.00 *gal*
Application event totals		0.00	0.00	0.00	42.70	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.44	5,702,410.00 *gal*
Application event totals		0.00	0.00	0.00	42.44	

3 - 11/28/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough Plant date: 11/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.09	8.52	87.81	489.56	486,795.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	28.35	4,015,053.00 *gal*
Application event totals		85.09	8.52	87.81	517.91	

03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	81.39	8.15	83.99	468.27	465,630.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	27.51	3,896,265.00 *gal*
Application event totals		81.39	8.15	83.99	495.79	

3 - 05/04/2023: Corn, silage

3 - 05/04/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 05/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.19	5,833,015.00 *gal*
Application event totals		0.00	0.00	0.00	41.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.72	5,766,352.00 *gal*
Application event totals		0.00	0.00	0.00	40.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.54	17.16	162.25	1,138.59	950,208.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.48	5,733,021.00 *gal*
Application event totals		90.54	17.16	162.25	1,179.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	126.79	23.35	220.41	1,483.98	1,019,898.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.01	5,666,358.00 *gal*
Application event totals		126.79	23.35	220.41	1,523.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	93.37	17.70	167.32	1,174.17	979,902.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.18	5,690,506.00 *gal*
Application event totals		93.37	17.70	167.32	1,214.36	

3 - 05/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.89	5,791,521.00 *gal*
Application event totals		0.00	0.00	0.00	40.89	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.66	5,757,849.00 *gal*
Application event totals		0.00	0.00	0.00	40.66	

4 - 11/25/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough

Plant date: 11/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	142.14	24.37	65.99	0.00	2,500.00 *ton*
Application event totals		142.14	24.37	65.99	0.00	

11/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	26.47	9,418,675.00 *gal*
Application event totals		0.00	0.00	0.00	26.47	

03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	26.73	9,511,470.00 *gal*
Application event totals		0.00	0.00	0.00	26.73	

4 - 05/04/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 05/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	321.24	55.07	149.15	0.00	5,650.00 *ton*
Application event totals		321.24	55.07	149.15	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.12	14,632,535.00 *gal*
Application event totals		0.00	0.00	0.00	41.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.84	14,532,541.00 *gal*
Application event totals		0.00	0.00	0.00	40.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.56	14,432,546.00 *gal*
Application event totals		0.00	0.00	0.00	40.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.09	14,265,888.00 *gal*
Application event totals		0.00	0.00	0.00	40.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.65	14,465,878.00 *gal*
Application event totals		0.00	0.00	0.00	40.65	

4 - 05/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.46	14,399,214.00 *gal*
Application event totals		0.00	0.00	0.00	40.46	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.93	14,565,872.00 *gal*
Application event totals		0.00	0.00	0.00	40.93	

5 - 11/27/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough

Plant date: 11/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	163.45	28.02	75.89	0.00	2,200.00 *ton*
Application event totals		163.45	28.02	75.89	0.00	

11/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	28.88	7,864,362.00 *gal*
Application event totals		0.00	0.00	0.00	28.88	

03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	28.62	7,794,766.00 *gal*
Application event totals		0.00	0.00	0.00	28.62	

5 - 05/05/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 05/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	338.03	57.95	156.94	0.00	4,550.00 *ton*
Application event totals		338.03	57.95	156.94	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.88	11,132,726.00 *gal*
Application event totals		0.00	0.00	0.00	40.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.61	11,332,715.00 *gal*
Application event totals		0.00	0.00	0.00	41.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.22	11,499,373.00 *gal*
Application event totals		0.00	0.00	0.00	42.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.61	11,332,715.00 *gal*
Application event totals		0.00	0.00	0.00	41.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.09	11,190,149.00 *gal*
Application event totals		0.00	0.00	0.00	41.09	

5 - 05/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.10	11,466,041.00 *gal*
Application event totals		0.00	0.00	0.00	42.10	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.35	11,532,704.00 *gal*
Application event totals		0.00	0.00	0.00	42.35	

Carpenter - 11/29/2022: Oats, silage-soft dough

Field name: Carpenter

Crop: Oats, silage-soft dough

Plant date: 11/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	150.02	25.72	69.65	0.00	7,000.00 *ton*
Application event totals		150.02	25.72	69.65	0.00	

11/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	30.60	28,892,732.00 *gal*
Application event totals		0.00	0.00	0.00	30.60	

03/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	30.40	28,700,435.00 *gal*
Application event totals		0.00	0.00	0.00	30.40	

Carpenter - 05/05/2023: Corn, silage

Field name: Carpenter

Crop: Corn, silage

Plant date: 05/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/30/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	330.03	56.58	153.23	0.00	15,400.00 *ton*
Application event totals		330.03	56.58	153.23	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.19	38,887,674.00 *gal*
Application event totals		0.00	0.00	0.00	41.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.88	38,593,813.00 *gal*
Application event totals		0.00	0.00	0.00	40.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.98	38,691,767.00 *gal*
Application event totals		0.00	0.00	0.00	40.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.27	40,856,122.00 *gal*
Application event totals		0.00	0.00	0.00	43.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.61	40,227,933.00 *gal*
Application event totals		0.00	0.00	0.00	42.61	

Carpenter - 05/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.54	40,159,923.00 *gal*
Application event totals		0.00	0.00	0.00	42.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.64	40,261,938.00 *gal*
Application event totals		0.00	0.00	0.00	42.64	

Crowslanding - 11/10/2019: Alfalfa, hay

Field name: Crowslanding

Crop: Alfalfa, hay

Plant date: 11/10/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	321.76	55.16	149.39	0.00	4,100.00 *ton*
Application event totals		321.76	55.16	149.39	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - River Water	Surface water	0.30	0.00	0.00	44.10	12,937,908.00 *gal*
Application event totals		0.30	0.00	0.00	44.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - River Water	Surface water	0.30	0.00	0.00	43.43	12,741,879.00 *gal*
Application event totals		0.30	0.00	0.00	43.43	

Crowslanding - 11/10/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - River Water	Surface water	0.30	0.00	0.00	43.10	12,643,865.00 *gal*
Application event totals		0.30	0.00	0.00	43.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - River Water	Surface water	0.30	0.00	0.00	43.77	12,839,894.00 *gal*
Application event totals		0.30	0.00	0.00	43.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - River Water	Surface water	0.31	0.00	0.00	44.53	13,064,593.00 *gal*
Application event totals		0.31	0.00	0.00	44.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - River Water	Surface water	0.31	0.00	0.00	44.88	13,165,608.00 *gal*
Application event totals		0.31	0.00	0.00	44.88	

Hogan South - 11/05/2022: Alfalfa, hay

Field name: Hogan South

Crop: Alfalfa, hay

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	122.58	21.01	56.91	0.00	3,300.00 *ton*
Application event totals		122.58	21.01	56.91	0.00	

Hogan South - 11/05/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - River Water	Surface water	0.27	0.00	0.00	39.14	24,258,578.00 *gal*
Application event totals		0.27	0.00	0.00	39.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - River Water	Surface water	0.27	0.00	0.00	39.30	24,357,592.00 *gal*
Application event totals		0.27	0.00	0.00	39.30	

Hogan South - 06/20/2023: Sudangrass, silage

Field name: Hogan South

Crop: Sudangrass, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	83.58	14.33	38.81	0.00	2,250.00 *ton*
Application event totals		83.58	14.33	38.81	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - River Water	Surface water	0.13	0.00	0.00	19.18	11,886,703.00 *gal*
Application event totals		0.13	0.00	0.00	19.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - River Water	Surface water	0.13	0.00	0.00	19.26	11,935,220.00 *gal*
Application event totals		0.13	0.00	0.00	19.26	

Hogan South - 06/20/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation water - River Water	Surface water	0.14	0.00	0.00	19.72	12,225,833.00 *gal*
Application event totals		0.14	0.00	0.00	19.72	

The 65 - 11/27/2022: Oats, silage-soft dough

Field name: The 65

Crop: Oats, silage-soft dough

Plant date: 11/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	145.73	24.98	67.66	0.00	1,700.00 *ton*
Application event totals		145.73	24.98	67.66	0.00	

11/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.13	8,527,621.00 *gal*
Application event totals		0.00	0.00	0.00	36.13	

03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.81	8,452,156.00 *gal*
Application event totals		0.00	0.00	0.00	35.81	

The 65 - 05/05/2023: Corn, silage

Field name: The 65

Crop: Corn, silage

Plant date: 05/05/2023

The 65 - 05/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	317.18	54.37	147.26	0.00	3,700.00 *ton*
Application event totals		317.18	54.37	147.26	0.00	

04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.65	11,246,325.00 *gal*
Application event totals		0.00	0.00	0.00	47.65	

06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.50	11,448,355.00 *gal*
Application event totals		0.00	0.00	0.00	48.50	

06/23/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.22	11,616,713.00 *gal*
Application event totals		0.00	0.00	0.00	49.22	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.48	11,679,635.00 *gal*
Application event totals		0.00	0.00	0.00	49.48	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.36	11,885,746.00 *gal*
Application event totals		0.00	0.00	0.00	50.36	

The 65 - 05/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.07	11,817,042.00 *gal*
Application event totals		0.00	0.00	0.00	50.07	

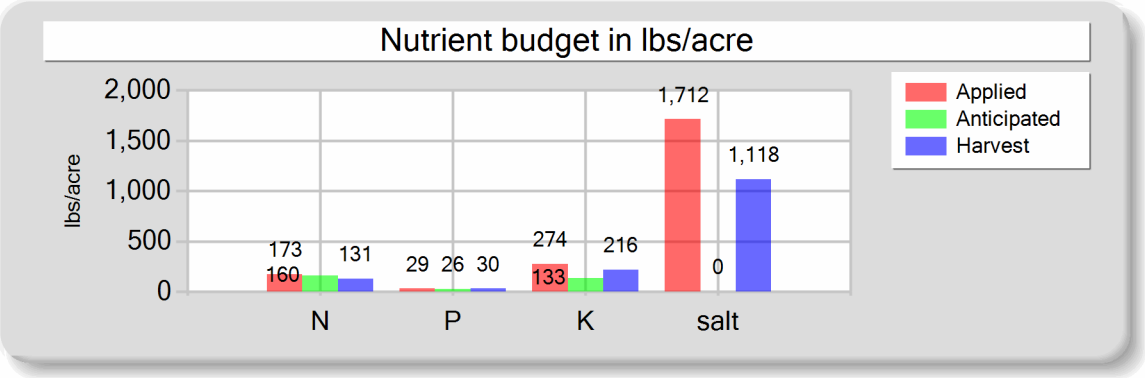
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.36	11,885,746.00 *gal*
Application event totals		0.00	0.00	0.00	50.36	

B. NUTRIENT BUDGET

1 - 11/28/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	165.92	29.32	274.15	1,635.84
Fresh water	0.00	0.00	0.00	75.96
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	172.92	29.32	274.15	1,711.80
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	130.61	29.89	215.84	1,117.97
Nutrient balance	42.31	-0.56	58.31	593.83
Applied to removed ratio	1.32	0.98	1.27	1.53

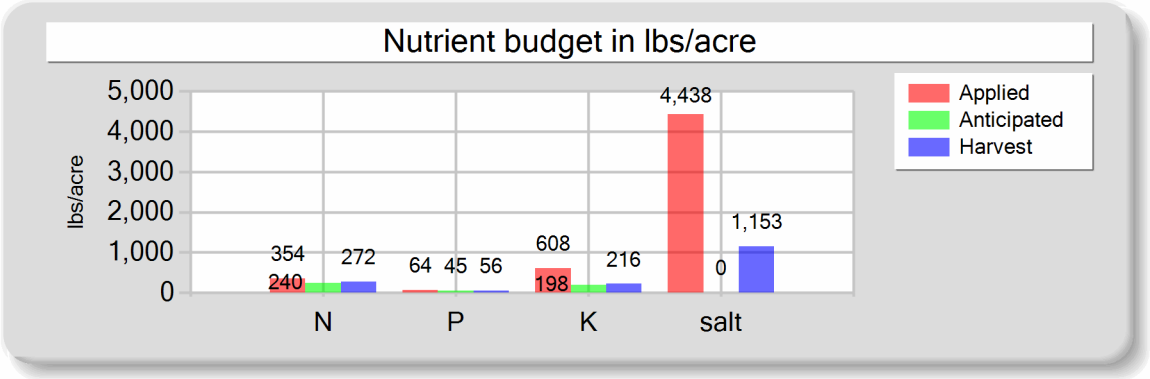
Fresh water applied	
10,206,044.00	*gallons*
375.85	*acre-inches*
10.16	*inches/acre*

Process wastewater applied	
1,209,431.00	*gallons*
44.54	*acre-inches*
1.20	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 05/04/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 05/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	347.21	64.43	608.45	4,142.71
Fresh water	0.00	0.00	0.00	295.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	354.21	64.43	608.45	4,437.87
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	271.69	55.57	216.12	1,152.64
Nutrient balance	82.52	8.86	392.33	3,285.22
Applied to removed ratio	1.30	1.16	2.82	3.85

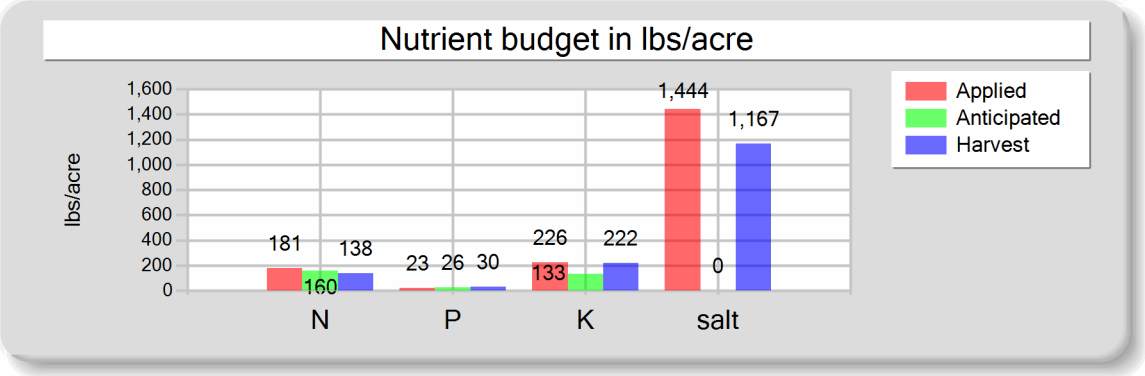
Fresh water applied	
39,656,939.00	*gallons*
1,460.43	*acre-inches*
39.47	*inches/acre*

Process wastewater applied	
2,860,284.00	*gallons*
105.33	*acre-inches*
2.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/28/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	173.78	22.90	225.56	1,389.20
Fresh water	0.00	0.00	0.00	54.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	180.78	22.90	225.56	1,444.05
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	138.38	29.74	221.87	1,166.55
Nutrient balance	42.39	-6.84	3.68	277.50
Applied to removed ratio	1.31	0.77	1.02	1.24

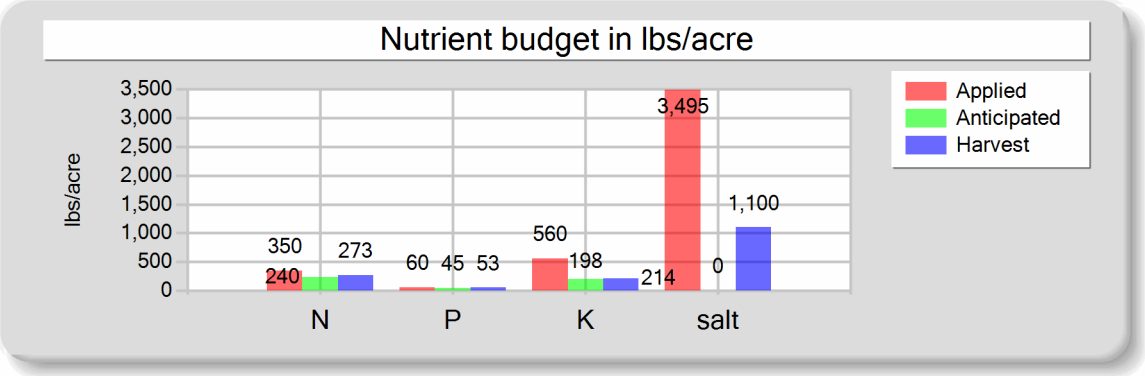
Fresh water applied	
7,370,043.00	*gallons*
271.41	*acre-inches*
7.34	*inches/acre*

Process wastewater applied	
1,087,396.00	*gallons*
40.05	*acre-inches*
1.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 05/04/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 05/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	343.16	60.09	559.82	3,203.48
Fresh water	0.00	0.00	0.00	291.81
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	350.16	60.09	559.82	3,495.29
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	273.10	53.05	214.16	1,100.25
Nutrient balance	77.06	7.04	345.66	2,395.04
Applied to removed ratio	1.28	1.13	2.61	3.18

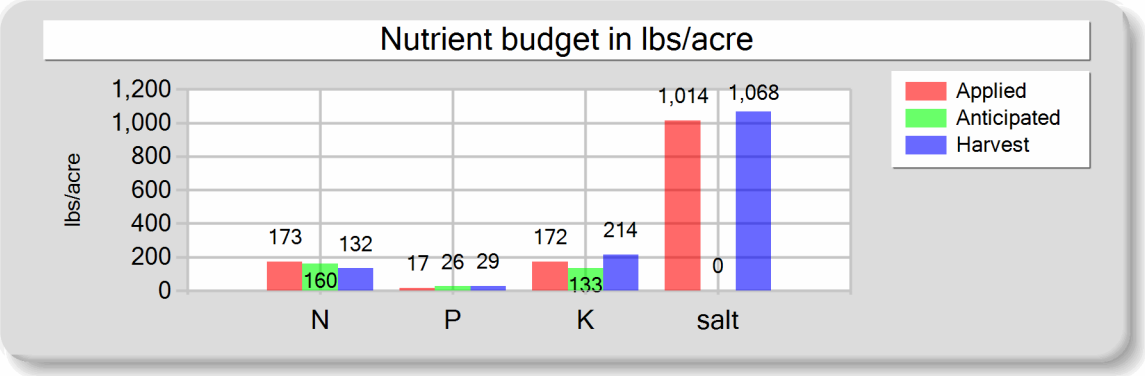
Fresh water applied	
39,207,364.00	*gallons*
1,443.87	*acre-inches*
39.02	*inches/acre*

Process wastewater applied	
2,667,558.00	*gallons*
98.24	*acre-inches*
2.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 11/28/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 11/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	166.48	16.67	171.80	957.83
Fresh water	0.00	0.00	0.00	55.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	173.48	16.67	171.80	1,013.70
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	132.22	28.56	213.66	1,068.32
Nutrient balance	41.26	-11.89	-41.86	-54.62
Applied to removed ratio	1.31	0.58	0.80	0.95

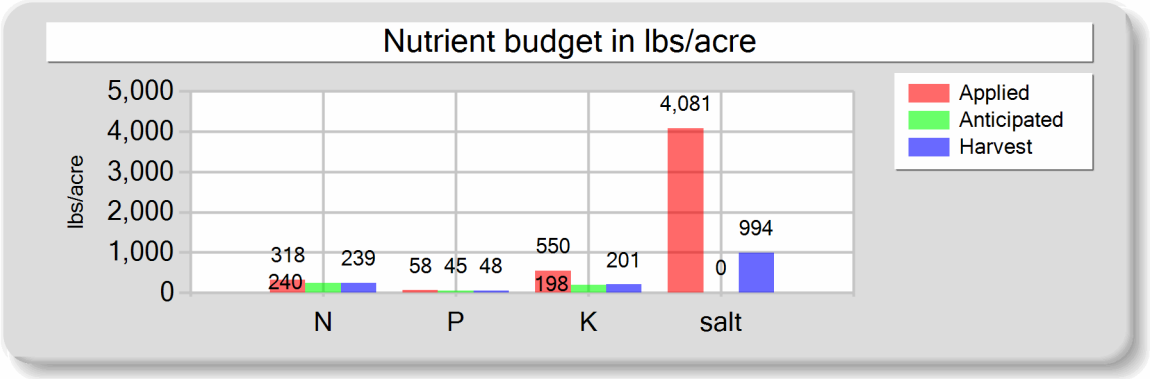
Fresh water applied	
7,911,318.00	*gallons*
291.35	*acre-inches*
7.47	*inches/acre*

Process wastewater applied	
952,425.00	*gallons*
35.07	*acre-inches*
0.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 05/04/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 05/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	310.70	58.21	549.98	3,796.74
Fresh water	0.00	0.00	0.00	284.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	317.70	58.21	549.98	4,080.88
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	239.38	47.88	200.71	994.36
Nutrient balance	78.32	10.33	349.27	3,086.52
Applied to removed ratio	1.33	1.22	2.74	4.10

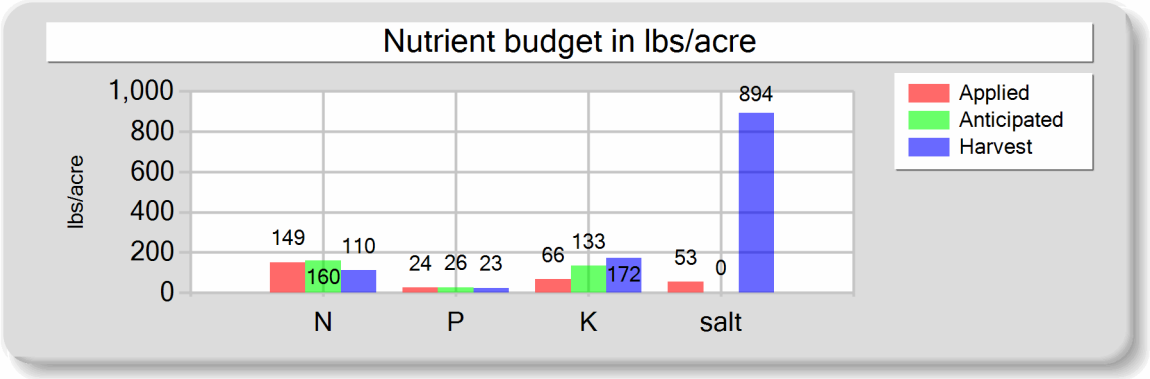
Fresh water applied	
40,238,622.00	*gallons*
1,481.85	*acre-inches*
38.00	*inches/acre*

Process wastewater applied	
2,950,008.00	*gallons*
108.64	*acre-inches*
2.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 11/25/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 11/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	142.14	24.37	65.99	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	53.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	149.14	24.37	65.99	53.19
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	110.37	22.80	172.39	893.88
Nutrient balance	38.78	1.56	-106.40	-840.68
Applied to removed ratio	1.35	1.07	0.38	0.06

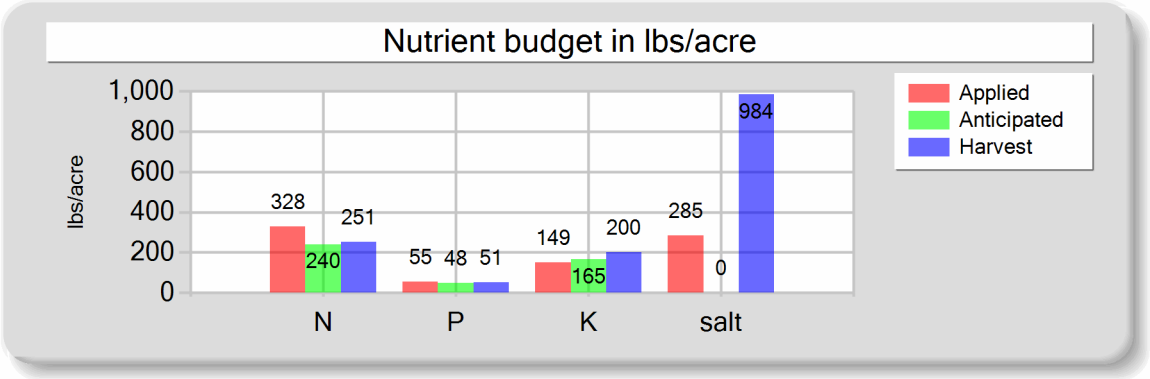
Fresh water applied	
18,930,145.00	*gallons*
697.13	*acre-inches*
7.11	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 05/04/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 05/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	321.24	55.07	149.15	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	284.64
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	328.24	55.07	149.15	284.64
Anticipated crop nutrient removal	240.00	48.00	165.00	0.00
Actual crop nutrient removal	251.46	51.02	200.44	983.96
Nutrient balance	76.79	4.05	-51.29	-699.32
Applied to removed ratio	1.31	1.08	0.74	0.29

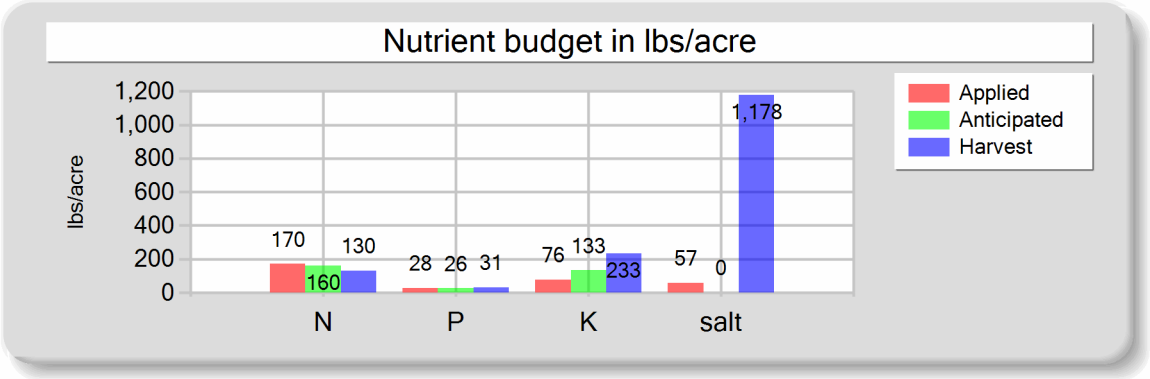
Fresh water applied	
101,294,474.00	*gallons*
3,730.33	*acre-inches*
38.06	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 11/27/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 11/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	163.45	28.02	75.89	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	57.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	170.45	28.02	75.89	57.50
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	129.55	30.62	233.20	1,177.76
Nutrient balance	40.89	-2.60	-157.31	-1,120.26
Applied to removed ratio	1.32	0.92	0.33	0.05

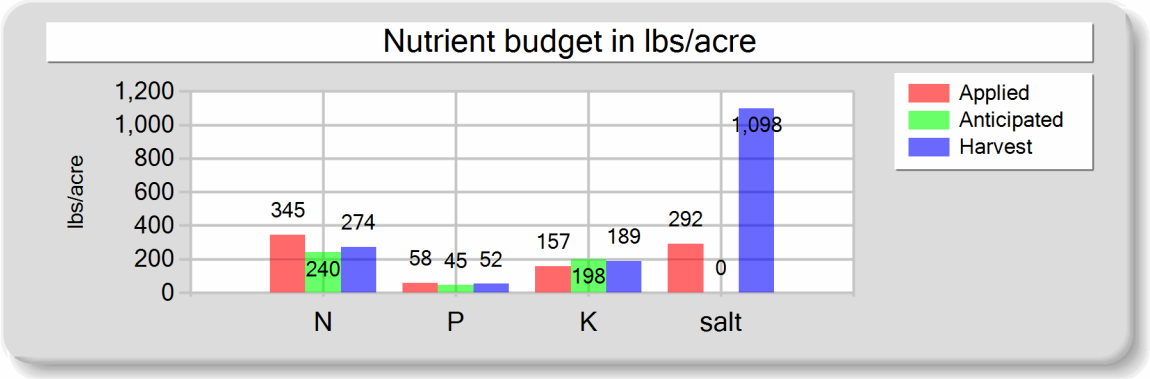
Fresh water applied	
15,659,128.00	*gallons*
576.67	*acre-inches*
7.69	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 05/05/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 05/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	338.03	57.95	156.94	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	291.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	345.03	57.95	156.94	291.86
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	273.54	52.01	188.78	1,098.02
Nutrient balance	71.49	5.94	-31.84	-806.16
Applied to removed ratio	1.26	1.11	0.83	0.27

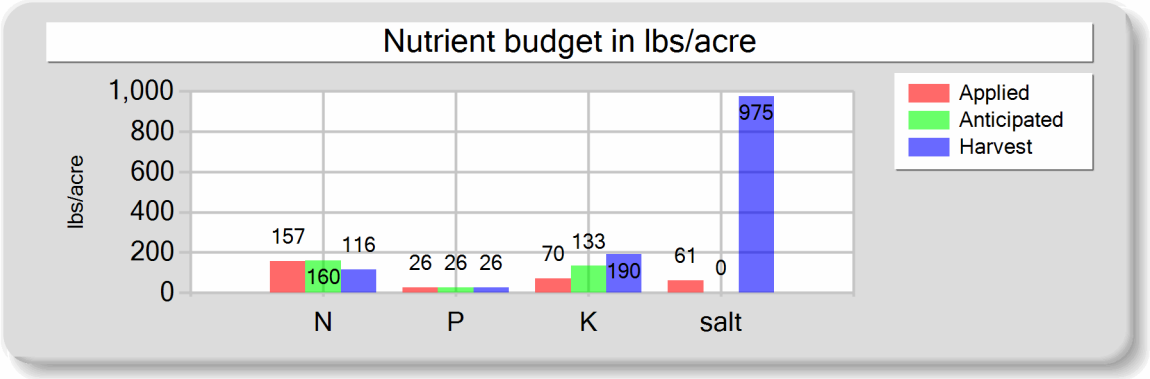
Fresh water applied	
79,486,423.00	*gallons*
2,927.21	*acre-inches*
39.03	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Carpenter - 11/29/2022: Oats, silage-soft dough

Field name: Carpenter Crop: Oats, silage-soft dough Plant date: 11/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	150.02	25.72	69.65	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	61.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	157.02	25.72	69.65	61.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	116.49	25.57	190.36	975.46
Nutrient balance	40.53	0.15	-120.71	-914.46
Applied to removed ratio	1.35	1.01	0.37	0.06

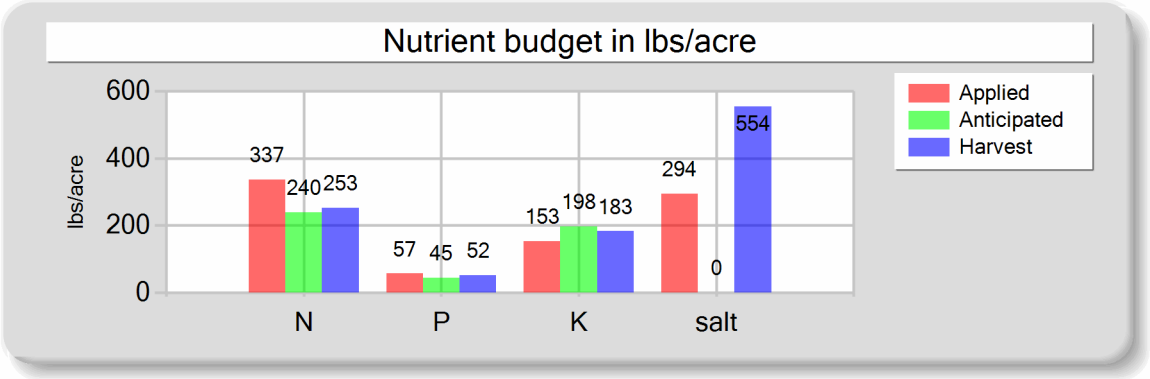
Fresh water applied	
57,593,167.00	*gallons*
2,120.96	*acre-inches*
8.16	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Carpenter - 05/05/2023: Corn, silage

Field name: Carpenter | Crop: Corn, silage | Plant date: 05/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	330.03	56.58	153.23	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	294.11
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	337.03	56.58	153.23	294.11
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	253.10	51.73	182.90	554.24
Nutrient balance	83.93	4.85	-29.67	-260.12
Applied to removed ratio	1.33	1.09	0.84	0.53

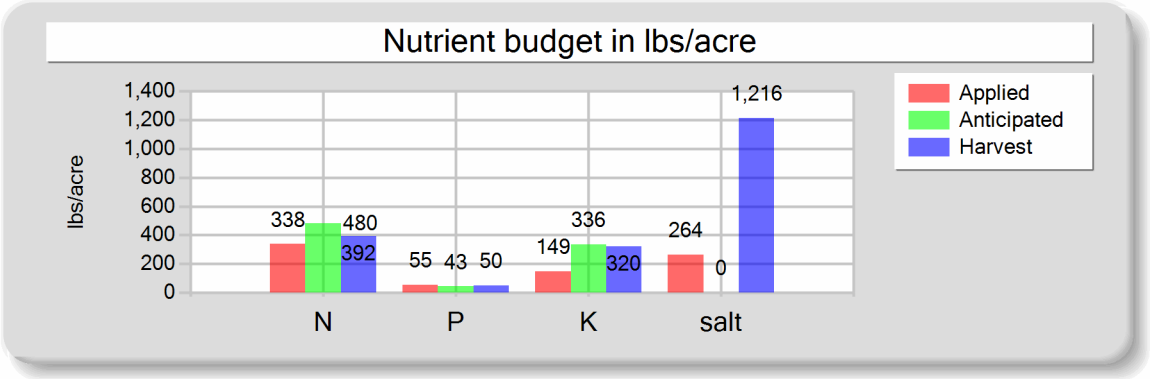
Fresh water applied	
277,679,170.00	*gallons*
10,225.98	*acre-inches*
39.33	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Crowslanding - 11/10/2019: Alfalfa, hay

Field name: Crowslanding Crop: Alfalfa, hay Plant date: 11/10/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	321.76	55.16	149.39	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.82	0.00	0.00	263.80
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	337.58	55.16	149.39	263.80
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	392.00	49.62	320.05	1,215.70
Nutrient balance	-54.42	5.54	-170.66	-951.90
Applied to removed ratio	0.86	1.11	0.47	0.22

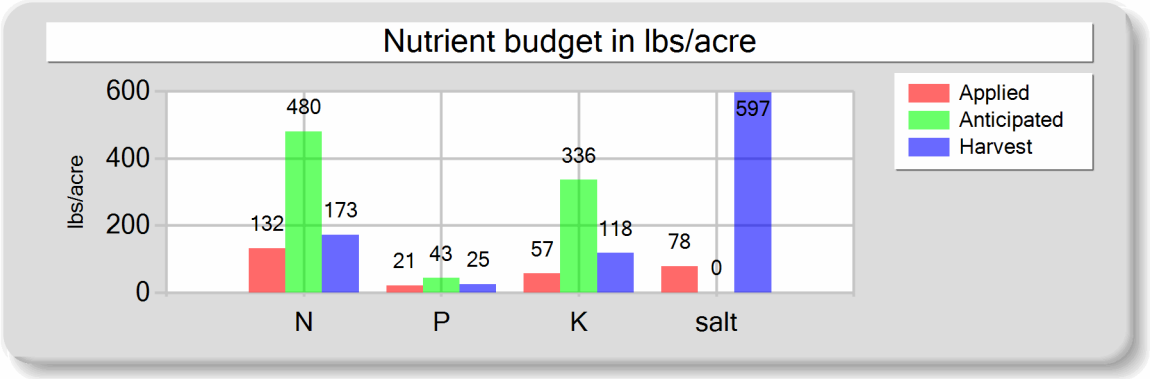
Fresh water applied	
77,393,747.00	*gallons*
2,850.15	*acre-inches*
40.14	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
6	*harvests*

Hogan South - 11/05/2022: Alfalfa, hay

Field name: Hogan South Crop: Alfalfa, hay Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	122.58	21.01	56.91	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.54	0.00	0.00	78.44
Atmospheric deposition	9.33	0.00	0.00	0.00
Total nutrients applied	132.46	21.01	56.91	78.44
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	172.85	24.90	118.08	597.49
Nutrient balance	-40.39	-3.88	-61.16	-519.05
Applied to removed ratio	0.77	0.84	0.48	0.13

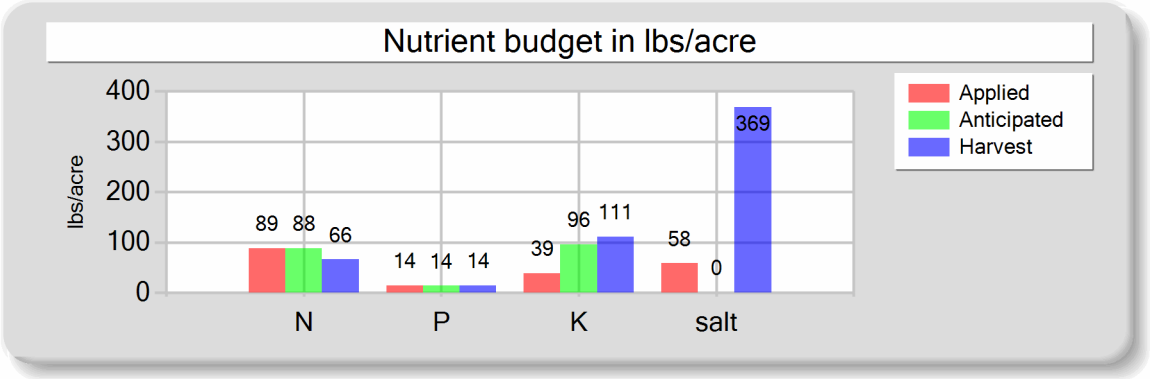
Fresh water applied	
48,616,170.00	*gallons*
1,790.37	*acre-inches*
11.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
2	*harvests*

Hogan South - 06/20/2023: Sudangrass, silage

Field name: Hogan South | Crop: Sudangrass, silage | Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	83.58	14.33	38.81	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.40	0.00	0.00	58.16
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	88.65	14.33	38.81	58.16
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	65.96	14.21	111.28	368.68
Nutrient balance	22.69	0.12	-72.48	-310.52
Applied to removed ratio	1.34	1.01	0.35	0.16

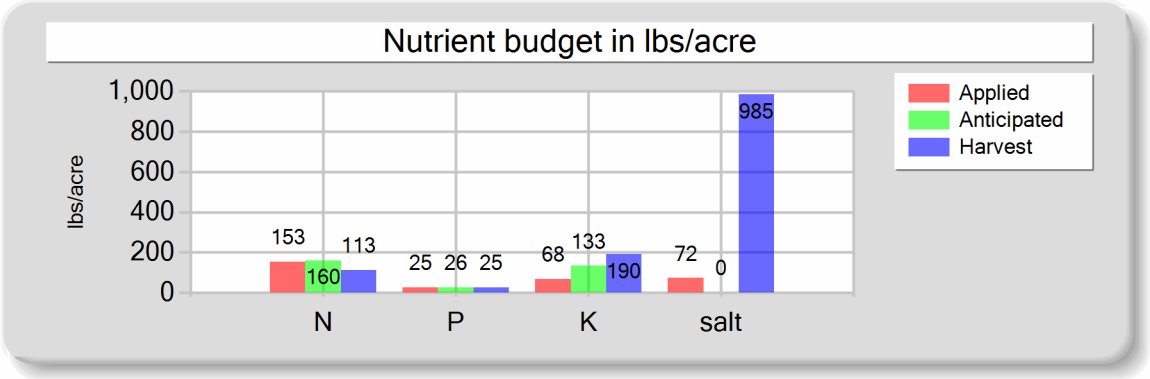
Fresh water applied	
36,047,756.00	*gallons*
1,327.52	*acre-inches*
8.85	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 65 - 11/27/2022: Oats, silage-soft dough

Field name: The 65 Crop: Oats, silage-soft dough Plant date: 11/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	145.73	24.98	67.66	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	71.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	152.73	24.98	67.66	71.94
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	112.94	25.10	190.17	984.62
Nutrient balance	39.79	-0.12	-122.51	-912.69
Applied to removed ratio	1.35	1.00	0.36	0.07

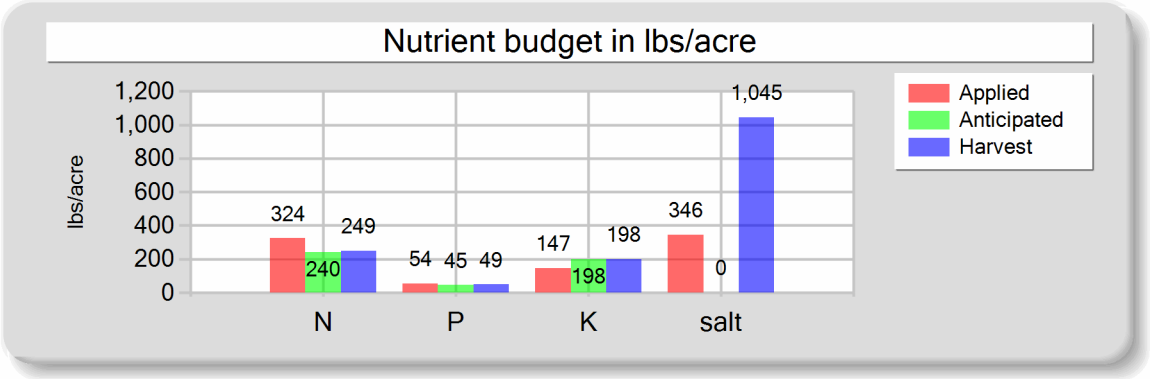
Fresh water applied	
16,979,777.00	*gallons*
625.31	*acre-inches*
9.62	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 65 - 05/05/2023: Corn, silage

Field name: The 65 Crop: Corn, silage Plant date: 05/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	317.18	54.37	147.26	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	345.63
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	324.18	54.37	147.26	345.63
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	248.75	48.67	198.28	1,045.46
Nutrient balance	75.43	5.70	-51.02	-699.84
Applied to removed ratio	1.30	1.12	0.74	0.33

Fresh water applied	
81,579,562.00	*gallons*
3,004.30	*acre-inches*
46.22	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure
Sample date: 05/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 78.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	1,200.00	2,300.00	4,900.00							0.00
DL	100.00	100.00	100.00							0.10

Solid Manure

Sample and source description: Solid Manure
Sample date: 10/16/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 80.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,000.00	2,400.00	6,500.00							0.00
DL	100.00	100.00	100.00							0.10

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water
Sample date: 02/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	815.00	373.00	0.00	1.90	81.80	843.00								8,800.00	4,700
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	635.00	439.00	0.00	1.00	108.00	1,000.00								9,990.00	5,180
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	445.00	316.00	0.00	0.30	84.40	798.00								77,750.00	5,600
DL	1.00	0.50	0.10	0.01	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	581.00	356.00	0.00	0.00	107.00	1,010.00								1,000.00	6,800
DL	1.00	0.50	1.00	1.00	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - DW-5627 Central

Domestic Well - DW-5627 Central

DW-5627 Central

Sample description: DW-5627 Central

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	13.90		13.90								1,680.00	0
DL	1.00		0.10								10.00	1

Domestic Well - DW-6025 Central

DW-6025 Central

Sample description: DW-6025 Central

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	25.70		25.70								2,250.00	0
DL	1.00		0.10								10.00	1

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 07/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00		0.00								32.90	33
DL	1.00		0.10								10.00	10

Irrigation water - River Water

Irrigation water - River Water

Irrigation water - River Water

Sample description: Irrigation water - River Water

Sample date: 07/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.20		0.20								37.80	29
DL	0.10		0.10								10.00	10

D. SOIL ANALYSES

1

Field 1

Sample and source description: Field 1

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			242.00				
DL			2.00				

2

Field 2

Sample and source description: Field 2

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			115.00				
DL			2.00				

3

3

Field 3

Sample and source description: Field 3

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			164.00				
DL			2.00				

4

Field 4

Sample and source description: Field 4

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			156.00				
DL			2.00				

5

Field 5

Sample and source description: Field 5

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			178.00				
DL			2.00				

Carpenter

Carpenter

Carpenter

Sample and source description: Carpenter

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			52.00				
DL			2.00				

Crowslanding

Crowslanding

Sample and source description: Crowslanding

Sample date: 01/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			25.00				
DL			2.00				

Hogan South

Hogan South-August

Sample and source description: Hogan South-August

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			45.00				
DL			2.00				

The 65

The 65

The 65

Sample and source description: The 65

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			58.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

1 - 11/28/2022: Oats, silage-soft dough

Field 1

Sample and source description: Field 1

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,700.00	19,500.00		10.10
DL	100.00	100.00	100.00		0.01

1 - 05/04/2023: Corn, silage

Field 1

Sample and source description: Field 1

Sample date: 08/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	2,700.00	10,500.00		5.60
DL	100.00	100.00	100.00		0.01

2 - 11/28/2022: Oats, silage-soft dough

Field 2

Sample and source description: Field 2

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,600.00	19,400.00		10.20
DL	100.00	100.00	100.00		0.01

2 - 05/04/2023: Corn, silage

Field 2

Sample and source description: Field 2

Sample date: 08/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	2,700.00	10,900.00		5.60
DL	100.00	100.00	100.00		0.01

3 - 11/28/2022: Oats, silage-soft dough

Field 3

Sample and source description: Field 3

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	2,700.00	20,200.00		10.10
DL	100.00	100.00	100.00		0.01

3 - 05/04/2023: Corn, silage

Field 3

Sample and source description: Field 3

Sample date: 08/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	2,600.00	10,900.00		5.40
DL	100.00	100.00	100.00		0.01

4 - 11/25/2022: Oats, silage-soft dough

Field 4 A

Sample and source description: Field 4 A

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,100.00	2,500.00	18,600.00		10.10
DL	100.00	100.00	100.00		0.01

Field 4 B

Sample and source description: Field 4 B

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,500.00	18,900.00		9.80
DL	100.00	100.00	100.00		0.01

4 - 05/04/2023: Corn, silage

4 - 05/04/2023: Corn, silage

Field 4 A

Sample and source description: Field 4 A

Sample date: 08/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,800.00	11,000.00		5.40
DL	100.00	100.00	100.00		0.01

Field 4 B

Sample and source description: Field 4 B

Sample date: 08/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,500.00	2,900.00	9,700.00		5.50
DL	100.00	100.00	100.00		0.01

5 - 11/27/2022: Oats, silage-soft dough

Field 5 A

Sample and source description: Field 5 A

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,000.00	2,600.00	19,400.00		10.20
DL	100.00	100.00	100.00		0.01

5 - 11/27/2022: Oats, silage-soft dough

Field 5 B

Sample and source description: Field 5 B

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,000.00	2,600.00	19,800.00		10.00
DL	100.00	100.00	100.00		0.01

5 - 05/05/2023: Corn, silage

Field 5

Sample and source description: Field 5

Sample date: 08/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,200.00	2,700.00	9,800.00		5.70
DL	100.00	100.00	100.00		0.01

Field 5 B

Sample and source description: Field 5 B

Sample date: 08/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	2,700.00	9,100.00		5.40
DL	100.00	100.00	100.00		0.01

Carpenter - 11/29/2022: Oats, silage-soft dough

Carpenter - 11/29/2022: Oats, silage-soft dough

Carpenter A

Sample and source description: Carpenter A

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,500.00	2,600.00	19,300.00		9.60
DL	100.00	100.00	100.00		0.01

Carpenter B

Sample and source description: Carpenter B

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,700.00	2,800.00	20,500.00		9.60
DL	100.00	100.00	100.00		0.01

Carpenter C

Sample and source description: Carpenter C

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	2,500.00	19,100.00		10.40
DL	100.00	100.00	100.00		0.01

Carpenter - 11/29/2022: Oats, silage-soft dough

Carpenter D

Sample and source description: Carpenter D

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	2,700.00	20,100.00		10.30
DL	100.00	100.00	100.00		0.01

Carpenter E

Sample and source description: Carpenter E

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,000.00	2,800.00	20,600.00		10.40
DL	100.00	100.00	100.00		0.01

Carpenter F

Sample and source description: Carpenter F

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,700.00	19,900.00		10.20
DL	100.00	100.00	100.00		0.01

Carpenter - 11/29/2022: Oats, silage-soft dough

Carpenter G

Sample and source description: Carpenter G

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,600.00	2,700.00	19,500.00		10.10
DL	100.00	100.00	100.00		0.01

Carpenter - 05/05/2023: Corn, silage

Carpenter C

Sample and source description: Carpenter C

Sample date: 08/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,100.00	2,500.00	10,200.00		5.70
DL	100.00	100.00	100.00		0.01

Carpenter A

Sample and source description: Carpenter A

Sample date: 08/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,600.00	2,800.00	10,600.00		5.30
DL	100.00	100.00	100.00		0.01

Carpenter - 05/05/2023: Corn, silage

Carpenter B

Sample and source description: Carpenter B

Sample date: 08/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,500.00	2,500.00	10,100.00		5.60
DL	100.00	100.00	100.00		0.01

Carpenter D

Sample and source description: Carpenter D

Sample date: 08/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	2,800.00	9,900.00		3.00
DL	100.00	100.00	100.00		0.01

Carpenter E

Sample and source description: Carpenter E

Sample date: 08/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,200.00	2,500.00	9,900.00		5.10
DL	100.00	100.00	100.00		0.01

Carpenter - 05/05/2023: Corn, silage

Carpenter F

Sample and source description: Carpenter F

Sample date: 08/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	2,400.00	10,200.00		5.70
DL	100.00	100.00	100.00		0.01

Carpenter G

Sample and source description: Carpenter G

Sample date: 08/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,200.00	2,600.00	10,100.00		5.80
DL	100.00	100.00	100.00		0.01

Crowslanding - 11/10/2019: Alfalfa, hay

Crowslanding A

Sample and source description: Crowslanding A

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 9.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,600.00	4,000.00	25,800.00		9.80
DL	100.00	100.00	100.00		0.01

Crowslanding - 11/10/2019: Alfalfa, hay

Crowslanding B

Sample and source description: Crowslanding B

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 4.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,000.00	3,400.00	11,400.00		8.00
DL	100.00	100.00	100.00		0.01

Crowlanding A

Sample and source description: Crowlanding A

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 9.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,200.00	3,000.00	16,800.00		8.00
DL	100.00	100.00	100.00		0.01

Crowslanding B

Sample and source description: Crowslanding B

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,500.00	3,200.00	17,400.00		8.50
DL	100.00	100.00	100.00		0.01

Hogan South - 11/05/2022: Alfalfa, hay

Hogan South - 11/05/2022: Alfalfa, hay

Hogan South-August A

Sample and source description: Hogan South-August A

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,900.00	3,200.00	11,000.00		8.30
DL	100.00	100.00	100.00		0.01

Hogan South-August B

Sample and source description: Hogan South-August B

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	36,100.00	3,900.00	17,300.00		10.30
DL	100.00	100.00	100.00		0.01

Hogan South-August C

Sample and source description: Hogan South-August C

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 5.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,300.00	3,500.00	16,600.00		8.40
DL	100.00	100.00	100.00		0.01

Hogan South - 11/05/2022: Alfalfa, hay

Hogan South-August D

Sample and source description: Hogan South-August D

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 5.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,400.00	3,600.00	11,800.00		8.30
DL	100.00	100.00	100.00		0.01

Hogan South - 06/20/2023: Sudangrass, silage

Hogan South-August A

Sample and source description: Hogan South-August A

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 82.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,200.00	4,400.00	29,600.00		11.30
DL	100.00	100.00	100.00		0.01

Hogan South-August B

Sample and source description: Hogan South-August B

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 83.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,000.00	4,400.00	28,900.00		10.80
DL	100.00	100.00	100.00		0.01

Hogan South - 06/20/2023: Sudangrass, silage

Hogan South-August C

Sample and source description: Hogan South-August C

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 82.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,100.00	4,400.00	32,000.00		11.30
DL	100.00	100.00	100.00		0.01

Hogan South-August D

Sample and source description: Hogan South-August D

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 83.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,500.00	4,200.00	32,900.00		10.90
DL	100.00	100.00	100.00		0.01

The 65 - 11/27/2022: Oats, silage-soft dough

The 65 A

Sample and source description: The 65 A

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,700.00	2,600.00	19,700.00		10.20
DL	100.00	100.00	100.00		0.01

The 65 - 11/27/2022: Oats, silage-soft dough

The 65 B

Sample and source description: The 65 B

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,500.00	2,700.00	20,000.00		10.10
DL	100.00	100.00	100.00		0.01

The 65 - 05/05/2023: Corn, silage

The 65 A

Sample and source description: The 65 A

Sample date: 08/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	2,700.00	10,000.00		5.60
DL	100.00	100.00	100.00		0.01

The 65 B

Sample and source description: The 65 B

Sample date: 08/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,700.00	11,000.00		5.80
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

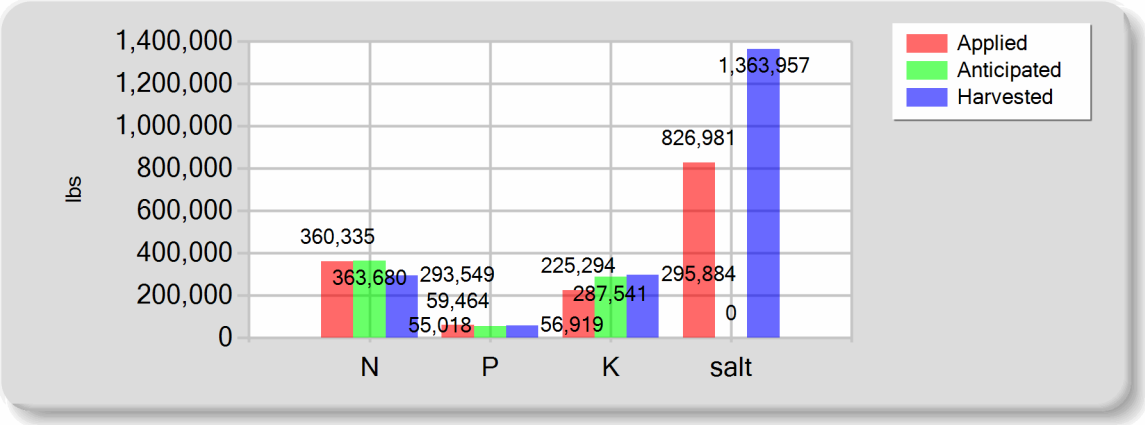
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

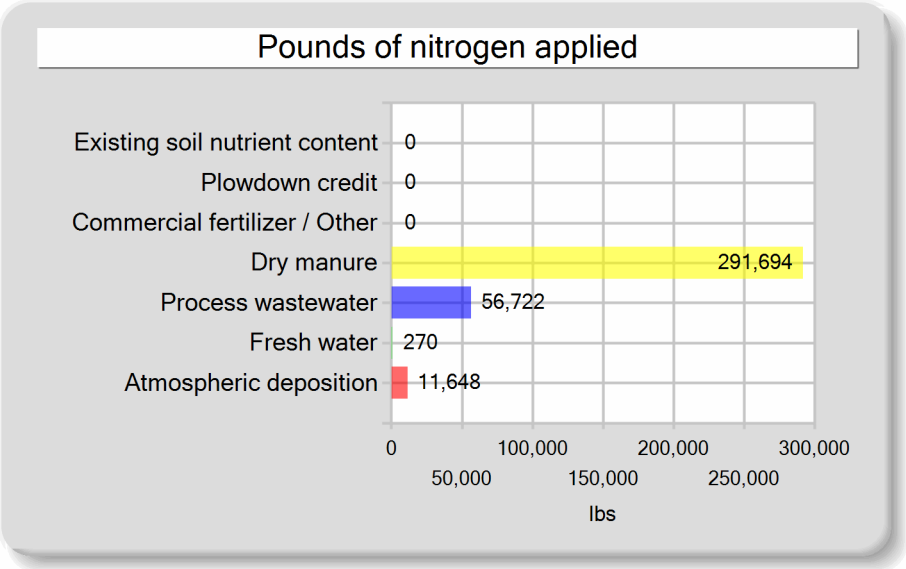
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

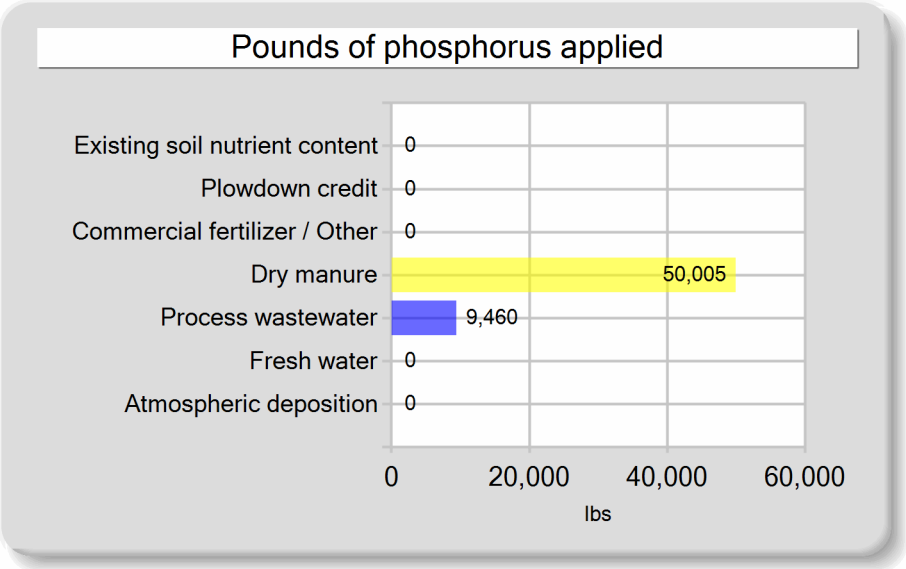
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	291,694.20	50,004.72	135,429.45	0.00
Process wastewater	56,722.41	9,459.74	89,864.38	569,163.56
Fresh water	270.47	0.00	0.00	257,817.23
Atmospheric deposition	11,648.00	0.00	0.00	0.00
Total nutrients applied	360,335.08	59,464.46	225,293.83	826,980.79
Anticipated crop nutrient removal	363,680.00	55,017.80	287,540.80	0.00
Actual crop nutrient removal	293,548.53	56,918.96	295,884.08	1,363,956.71
Nutrient balance	66,786.55	2,545.50	-70,590.25	-536,975.92
Applied to removed ratio	1.23	1.04	0.76	0.61

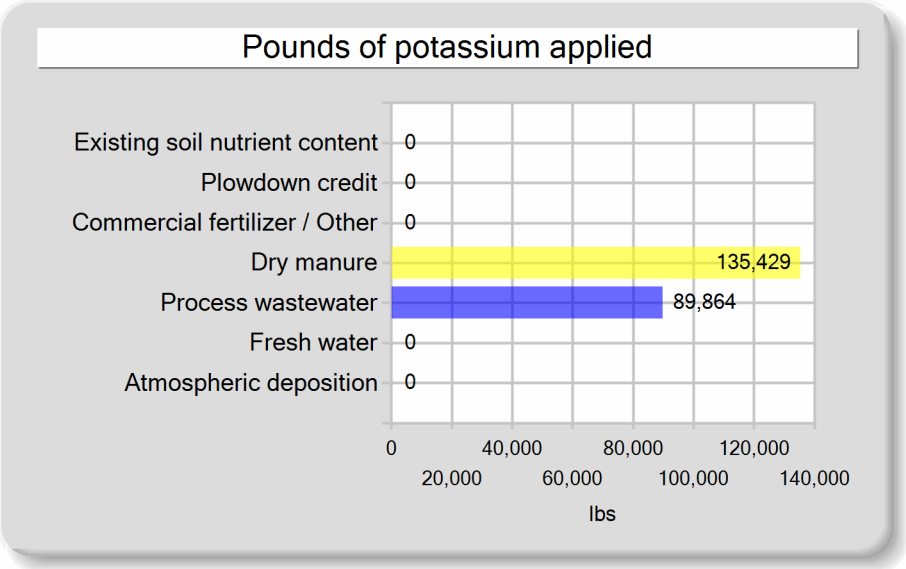
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

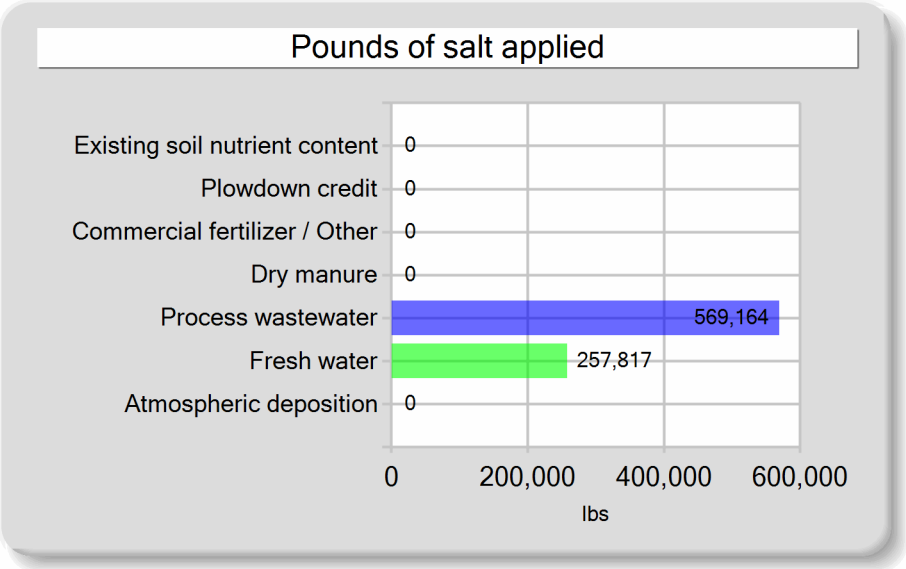


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Joe Jordao (Jun 20, 2024 15:04 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Joe Jordao	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/20/2024	
DATE	DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Jordao Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:21

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0670-01	DW-5627 Central	Ag Water	T. Madruga		11/13/2023 15:00
23K0670-02	DW-6025 Central	Ag Water	T. Madruga		11/13/2023 16:00

Default Cooler Temperature on Receipt °C: 4.6
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Ranch: Jordao Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:21

Sample Results

Sample: DW-5627 Central
23K0670-01 (Water)

Sampled: 11/13/2023 15:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.68**	mmhos/cm	0.01	1		11/16/23 12:48	SM 2510 B		BEK0477
Electrical Conductivity umhos	**1680**	umhos/cm	10.0	1		11/16/23 12:48	SM 2510 B		BEK0477
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 15:00	Field		BEK0417
Nitrate Nitrogen as NO3N	**13.9**	mg/L	0.1	1	10	11/14/23 21:01	EPA 300.0		BEK0412
Temperature	**25.0**	units	0.0	1		11/16/23 12:48	SM 4500-H+	H	BEK0477
pH	**7.7**	units	1.0	1		11/16/23 12:48	SM 4500-H+	H	BEK0477

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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Jordao Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:21

Sample Results
(Continued)

Sample: DW-6025 Central
23K0670-02 (Water)

Sampled: 11/13/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.25**	mmhos/cm	0.01	1		11/16/23 12:49	SM 2510 B		BEK0477
Electrical Conductivity umhos	**2250**	umhos/cm	10.0	1		11/16/23 12:49	SM 2510 B		BEK0477
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 15:00	Field		BEK0417
Nitrate Nitrogen as NO3N	**25.7**	mg/L	0.1	1	10	11/14/23 21:20	EPA 300.0		BEK0412
Temperature	**25.0**	units	0.0	1		11/16/23 12:49	SM 4500-H+	H	BEK0477
pH	**7.6**	units	1.0	1		11/16/23 12:49	SM 4500-H+	H	BEK0477

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LABORATORY INC

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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Jordao Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:21

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0412										
Blank (BEK0412-BLK1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0412-BLK2)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0412-BLK3)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0412-BS1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		100	90-110		
LCS (BEK0412-BS2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		100	90-110		
Duplicate (BEK0412-DUP1)		**Source: 23K0676-01**			Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	15.7		0.1	mg/L		15.5			0.802	10
Duplicate (BEK0412-DUP2)		**Source: 23K0676-02**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	36.4		0.1	mg/L		36.4			0.0192	10
Matrix Spike (BEK0412-MS1)		**Source: 23K0676-01**			Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	20.6		0.1	mg/L	5.000	15.5	102	90-110		
Matrix Spike (BEK0412-MS2)		**Source: 23K0676-02**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	42.1		0.1	mg/L	5.000	36.4	113	90-110		
Reference (BEK0412-SRM1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		97.9	90-110		
Reference (BEK0412-SRM2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		98.7	90-110		
Reference (BEK0412-SRM3)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		99.2	90-110		

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DELLAVALLE™ LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Jordao Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:21

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0477										
Blank (BEK0477-BLK1)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.5		1.0	units						
Blank (BEK0477-BLK2)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.5		1.0	units						
Blank (BEK0477-BLK3)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.3		1.0	units						
Duplicate (BEK0477-DUP1)		**Source: 23K0667-01**			Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	4.04		0.01	mmhos/cm		4.04			0.0495	10
pH	7.8		1.0	units		7.6			1.82	10
Electrical Conductivity umhos	4040		10.0	umhos/cm		4040			0.0495	10
Duplicate (BEK0477-DUP2)		**Source: 23K0676-03**			Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	1.35		0.01	mmhos/cm		1.36			0.791	10
Electrical Conductivity umhos	1350		10.0	umhos/cm		1360			0.791	10
pH	7.6		1.0	units		7.6			0.264	10
Reference (BEK0477-SRM1)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	578			umhos/cm	538.0		107	90-110		
Reference (BEK0477-SRM2)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEK0477-SRM3)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0477-SRM4)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEK0477-SRM5)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
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Account# 00-0019840
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Submitted By: Rui & John Avila
Ranch: Jordao Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:21

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0477 (Continued)										
Reference (BEK0477-SRM5)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0477-SRM6)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0477-SRM7)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0477-SRM8)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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11/14/23 15:45 23K0670

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples 2 No. of Bottles: 2

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Results Need By

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT Jordao Dairy
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:
Analysis and Bottles Required: (Please indicate Analysis)
(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	DW-5627 Central	11/13/2023	1500	4.6	ND
2	DW-6025 Central	11/13/2023	1600	3.6	ND
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/13/2023 @ 1600	11/14/2023 @ 900
Second	JLF	Dellavalle Laboratory, Inc.	11/14/2023 @ 900	11-14-23 3:30
Third				
Fourth		DCL	11/14 15:45	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/14/23 15:45 23K0670

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	\	\								
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										